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Cantilever Arm Design Proposal

Prepared for: [Client]
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Major Deliverables

- Appropriate aluminum beam thickness where first bending natural frequency >= 75 Hz to avoid resonance:
  - Minimum beam thickness: **44.384 mm**
  - McMaster Carr stock part number: **6945T812**
- Recommended methodology for future analysis:
  - Built-in MATLAB **fzero function** for the best balance of reliability and efficiency

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Our team at Burdell Inc. came to a few specific recommendations:

- First, we determined that an aluminum beam with a minimum thickness of 44.384 mm ensures a first natural bending frequency >= 75 Hz, which will prevent resonance from the gearbox that threatens excessive oscillations, jarring vibrations, or even catastrophic structural failure.
- Next, we have identified a commercially viable stock aluminum beam available on McMaster Carr, part number 6945T812, matching base dimensions and slightly exceeding the minimum required thickness by 6mm, ensuring effectiveness for the target.
- Finally, for future analyses you may require, we recommend using MATLAB's fzero root finding function, as it provides a balance of robustness and efficiency in terms of operating speed.

Outline

1. Find ideal minimum thickness
   - Root finding approach
   - Derive frequency equation
   - Plot first bending natural frequency vs. beam thickness
   - Mass variation study

2. Convergence Performance Analysis
   - Plot runtime and iterations vs. beam thickness
   - Evaluate failure cases of each method
   - Analysis and Conclusions

3. Conclusions and Recommendations

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Bending Frequency

Solving roots numerically and analyzing frequency equation

Performance analysis

Evaluating effectiveness of root finding methods

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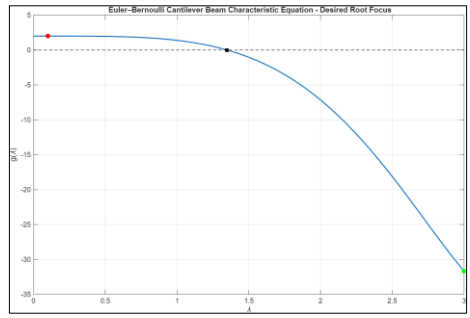
Root Finding Method

Modeling the link as a **Euler-Bernoulli** cantilever beam with a lumped tip mass gives a characteristic equation that cannot be solved algebraically.

- The bending behavior is governed by $\boldsymbol{\lambda}$
- We must recast the implicit characteristic equation as a residual
- We need the lowest valued real root where $\boldsymbol{g(\lambda) = 0}$ to find the first bending mode
- We must solve for $\boldsymbol{\lambda}$ numerically, using root-finding algorithms such as **bisection**, **Newton-Raphson**, or MATLAB's **fzero** function



$$1 + \cosh\lambda\cos\lambda = -\mu\lambda(\sinh\lambda\cos\lambda - \cosh\lambda\sin\lambda)$$

$$\boldsymbol{R(\lambda) = 1 + \cosh(\lambda)\cos(\lambda) + \mu\lambda(\sinh(\lambda)\cos(\lambda) - \cosh(\lambda)\sin(\lambda))}$$

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To determine and analyze the first bending natural frequency of the beam, we must start by finding the first bending mode, which is represented by the first real root of $\boldsymbol{\lambda}$. This describes the fundamental way the beam tends to bend and vibrate in relation to its static base. The higher roots give more complex information about the bending behavior around several nodes along the beam's length with higher resonance frequencies, but only the first root is relevant for our conditions.

Due to hyperbolic trigonometric terms, the characteristic equation cannot be solved algebraically, so numerical root finding methods are the only valid approach, which we can achieve with the help of coded MATLAB functions to speed up the process. Fortunately, the characteristic equation is continuous and differentiable so we can use the following methods:

- Bisection picks an x-interval around the root and cuts in half until it converges onto the root for an accurate approximation. This is a robust and simple method, however it requires the root to be inside the interval and could need many iterations to succeed.
- Newton-Raphson uses an initial x-value guess close to the root and uses the tangent (derivative) line's x-intercept as the next guess until it converges

quadratically on the answer. The initial guess must be close to the root or else the method will give the wrong root or diverge.

- Fzero is a built-in MATLAB function that uses a combination of root finding methods to give an accurate approximation. It omits the use of any derivatives which keeps it simple and well-optimized.

We will later analyze each method's effectiveness to see which is most viable for the applications of the project.

Bending Frequency Equation

With $\boldsymbol{\lambda}$ found from numerical methods, we now must derive the first bending frequency equation

- Convert angular frequency $\boldsymbol{\omega}$ to natural frequency $\boldsymbol{f}$
- Convert area and moment of inertia into terms of $\boldsymbol{t}$ so we can solve for **first bending natural frequency as a function of thickness**
- Reapproximate $\boldsymbol{\lambda}$ for each step of thickness
- This function can be used to determine a minimum thickness that avoids resonance

$$\omega = \beta^2 \sqrt{\frac{EI}{\rho A}} = \frac{\lambda^2}{L^2} \sqrt{\frac{EI}{\rho A}}$$

$$\Downarrow$$

$$f_1 = \frac{\omega}{2\pi} = \frac{\lambda^2}{2\pi L^2} \sqrt{\frac{EI}{\rho A}}$$

$$\Downarrow$$

$$f_1(t) = \frac{\lambda^2 t}{4\pi L^2 \sqrt{3}} \sqrt{\frac{E}{\rho}}$$

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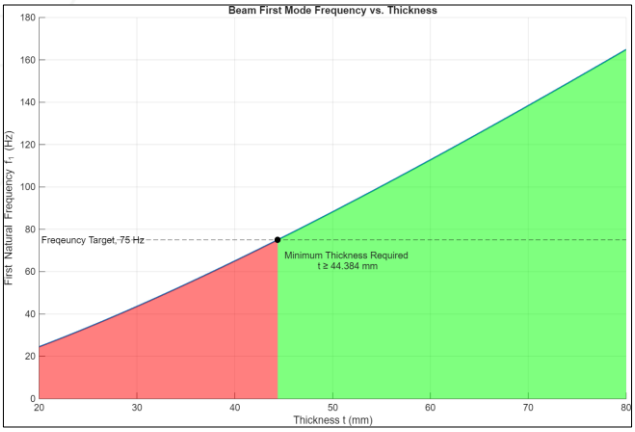
We can use known Euler-Bernoulli equations to compute first bending natural frequency as a function of thickness.

- We can derive the equation through the steps shown, using known equations and constants provided by the client.
- First bending natural frequency depends on lambda, which we now know how to find numerically with root finding methods.
- It is important to note that lambda is not a constant, but rather a function of thickness because the mass ratio mu changes with the beam's cross-sectional area. Consequently, as thickness changes by a given step, lambda must be reapproximated by the root finding methods coded in MATLAB and substituted back into the frequency function.
- Thus, the runtime and number of iterations for root finding are an important factor to weigh in for the optimization of the method, which will be evaluated in the performance analysis.
- The final equation can be plotted to analyze the relationship between frequency and thickness, which we can use to find the minimum thickness to avoid resonance.

Proposed Minimum Thickness



Bending frequency and thickness have a direct relationship

- The minimum thickness to avoid resonance lies where the curve crosses 75 Hz
- Any thickness >= 44.384 mm yields a first bending frequency >= 75 Hz
- McMaster-Carr stock 6945T812 part is recommended
  - 30mm x 2" x 24" satisfies pre-requisite dimensions and minimum thickness (2" > 44.428mm)

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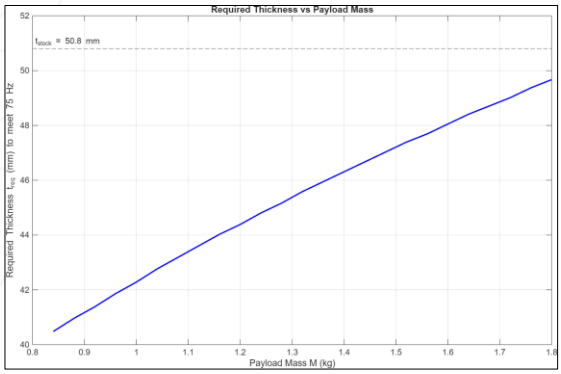
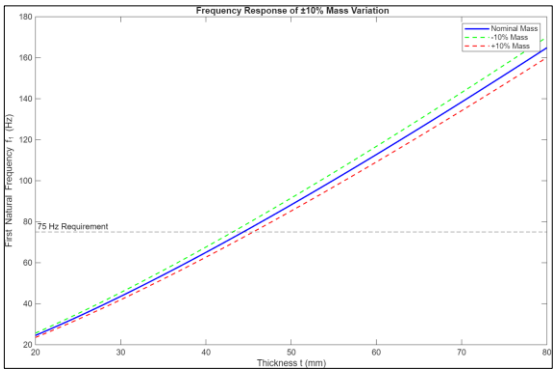
Analyzing the blue curve, we can clearly see that as thickness increases, first bending natural frequency also increases. We must evaluate the minimum thickness where frequency equals 75 Hz, determined from 1.25 (for a safety factor) times the gearbox's 60Hz excitation. While the graph may look almost linear, the change of lambda gives it a slight curve so we can't just take a slope. MATLAB tells us that the line crosses 75 Hz at a minimum thickness of 44.384 millimeters, which aligns with the information on the plot. Thus, the client must choose a thickness greater than or equal to this value to soundly achieve the goal of preventing resonance from a 60 Hz torque ripple from the gearbox. At any lower thickness, the beam is at risk of vibrations from the gearbox, and the problem will persist. Burdell Inc. recommends using the given McMaster-Carr stock Aluminum beam as it aligns with the client's requested material and dimensions, and it exceeds the minimum thickness, at 2", or 50.8mm, thick. This stock guarantees the safety margin without redesign, and is a readily accessible part at a reasonable price, and can be ordered in bulk without custom manufacturing charges.

Further Analysis – Stock Justification





- If tooling mass changes in future revisions, the same method is valid
  - The selected stock has a thickness of 2" = 50.8mm (> required 44.438mm)
  - Payload can vary by >10% and remain within the respective required thickness for the 75Hz resonance avoidance
  - This recommended material therefore provides exemplary performance

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Although the tooling mass is currently fixed for this study, we also evaluated how the beam would perform under different payloads, since the recommended stock's thickness is greater than required.

We found that even with more than a 10% increase in payload mass, the required thickness to maintain at least a 75 Hz natural frequency remains below the 50.8mm of the recommended stock's thickness. This is significant because it confirms that the selected 2" stock provides compliance for your current needs, as well as robustness for future equipment changes that can potentially lessen redesign.

This analysis quantifies that value of selecting a slightly thicker stock, even though it is not explicitly needed for the current use, which includes a safety factor for the Hz requirement already. This provides security in performance for future analysis of your product as potential redesigns, configurations, and other changes occur.

Bending frequency

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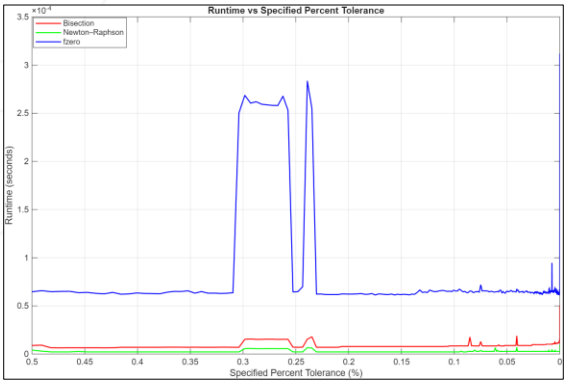
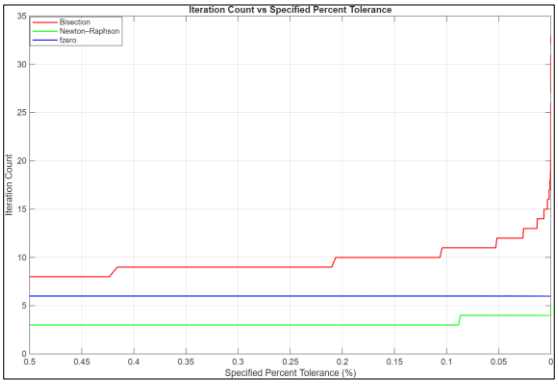
Iterations and Runtime vs. Percent Tolerance



Runtime vs Specified Percent Tolerance



Iteration vs Specified Percent Tolerance

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We must ensure the root finding methods are well optimized for an accurate level of convergence on the true value of the root. The root finding methods run many times throughout the plotting process, so they play an important role in the accuracy of results and they must do so at a reasonable efficiency. We can evaluate this by considering the number of iterations as well as runtime as functions of percent tolerance. Along the x-axes, the upper bound shows more accurate convergence with 10 significant figures of agreement while the lower bound shows a more flexible specification with 2 significant figures of agreement.

As seen on the left, each of the methods drastically increase in runtime as percent tolerance reaches 0. This reflects that these are approximations and getting a direct result is virtually impossible. Nevertheless, each of the methods can produce a highly accurate result with error reaching effectively zero in a reasonable amount of time.

Fzero has a noticeably higher runtime than other methods, while its iterations are reasonable.

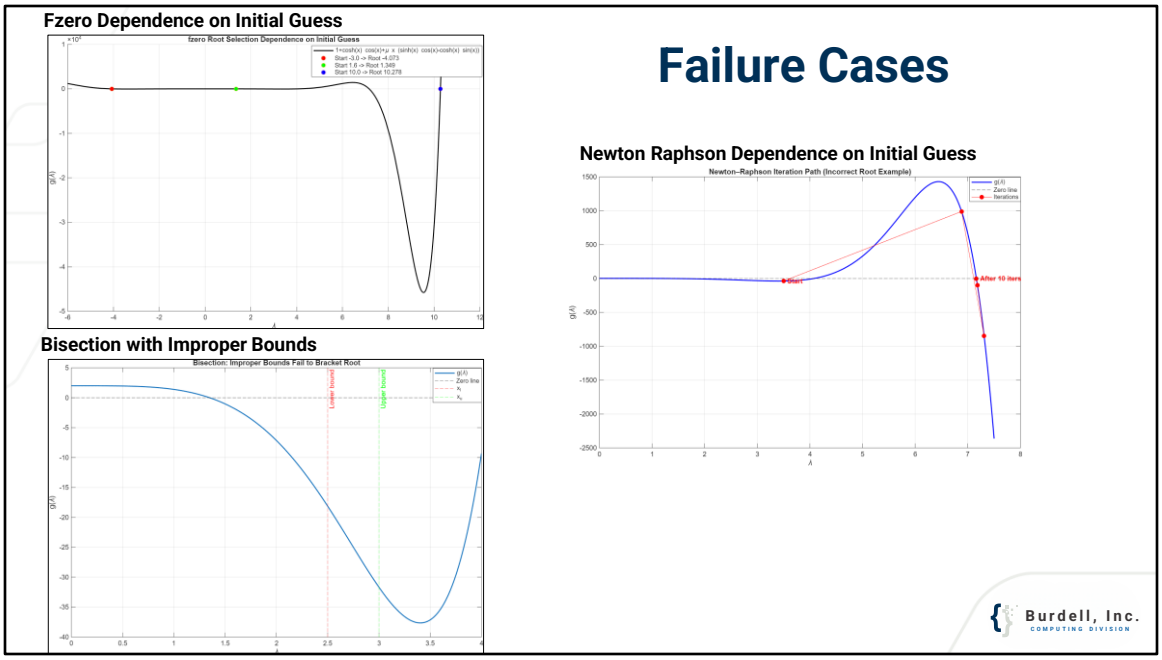
Newton Raphson appears very robust with the lowest runtime and lowest iterations.

Bisection uses exponentially more iterations as percent tolerance gets more

stringent but regardless has a low runtime.

By comparing iteration count and runtime versus percent tolerances, we identify how each method behaves as tighter accuracy is demanded to find potential inefficiencies or failure behaviors. This informs us on the method choice, and on overconstraining tolerance. Sensitive algorithms such as Newton raphson may not need extremely high tolerances, as they provide no additional gain.

Overall, all of these methods run very quickly, with the differences noticeable to a human, and can provide a very high degree of tolerance. Use cases vary based on the function and demands, which we'll get into for your case next.



While all three methods successfully located the desired root in our analysis in a rapid amount of time, <1ms, all can fail if applied incorrectly, which is critical to consider in future projects.

Bisection is only reliable if you know the true root is between the initial bounds you select. If the guesses bracket the wrong interval, the method will either not be able to converge, or it will converge to the wrong root, giving an entirely different frequency

Newton-Raphson is very efficient, but also the most sensitive. Unnoticeable changes to the curve of the function can cause divergence, or a jump to a much further root. This can dangerously lead an analyst to believe a design is safe when it is not, or the other way around. This can also locate a distant root incorrectly.

Fzero is a very reliable and consistent function, but like the others, can locate a root beyond the first one if the initial guess is not informed, as this method will converge to the root that is closest in all cases, and is therefore the most dependable and predictable

All of these methods can be susceptible to failure, so understand these failure cases for your future analyses.

Analysis and Conclusions

NEWTON-RAPHSON		FZERO	
Initial Guess	Root found	Initial Guess	Root found
0	0	0	-1.3493
0.20408	95.041	0.20408	1.3493
0.40816	10.278	0.40816	1.3493
0.61224	4.0735	0.61224	1.3493
0.81633	1.3493	0.81633	1.3493
1.0204	1.3493	1.0204	1.3493
2.8571	1.3493	2.8571	4.0735
3.0612	1.3493	3.0612	4.0735
3.2653	148.45	3.2653	4.0735
3.4694	7.1632	3.4694	4.0735
3.6735	4.0735	3.6735	7.1632
3.8776	4.0735	3.8776	7.1632
5.9184	4.0735	5.9184	7.1632
6.1224	4.0735	6.1224	7.1632
6.3265	1.3493	6.3265	7.1632
6.5306	10.278	6.5306	7.1632
6.7347	7.1632	6.7347	7.1632
6.9388	7.1632	6.9388	7.1632
8.9796	7.1632	8.9796	10.278
9.1837	7.1632	9.1837	10.278
9.3878	4.0735	9.3878	10.278
9.5918	16.537	9.5918	10.278
9.7959	10.278	9.7959	10.278
10	10.278	10	10.278

Fzero is the most reliable in finding the closest root

- Fzero shows clear consistency in its root finding results
- Newton-Raphson has occasional cases that land on roots far away from the initial guess

We recommend Fzero for its balance between reliability and efficiency. With any method used, evaluate behavior near the root to determine initial guesses.

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Fzero has the best balance between reliability and efficiency. The data shown in the table shows that fzero tends to converge on the nearest root, so it only relies on the user to make a reasonable guess. Newton-Raphson similarly relies on an initial guess but sometimes displays roots far from the initial guess rather than the nearest root, which could cause serious discrepancies in results. Thus, it is not an adequate method that can be trusted for the purposes of the project. Bisection requires two guesses, which leaves more room for error and failures in the method. Using fzero ensures peace of mind when producing results to determine the minimum thickness, and its slightly slower runtime is hardly a sacrifice when considering its accuracy.

The most important consideration when using any of these numeric methods, however, is to use informed guesses that hinge on knowledge of the behavior of the function around the desired root whether that be the first, second, or so on. Closer initial guesses, when possible, are informed starting values that both lessen the computing load and ensure that the correct solution is found, and product goals are met.

This is why fzero is recommended for its steadily predictable convergence over a

range of guesses. It supports accurate design decisions with minimal risk involved.

We recommend:

- The purchase and use of McMaster-Carr stock **6945T812** to provide sufficient thickness margin
- Future analyses to be done with MATLAB's fzero function, with informed initial guesses

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To safely avoid resonance from the 60 Hz torque ripple from the gearbox, Burdell Inc. recommends:

- Use an aluminum bar of minimum thickness 44.384 mm for a first bending natural frequency of 75 Hz that will prevent unwanted resonance.
- Order McMaster Carr stock Aluminum 6016 bar 6945T812 – it aligns with project specifications, guarantees security against resonance, and is readily available for a good price.

  By doing so, we assure you that the resosaance risk will safely be eliminated. With this solution design, and the validated methods we've shown you, we provide you a solution that is safe both today, and a validated method you can evolve alongside your system.

Thank you for your business!